

ADDITIONAL PROBLEM SHEET 3: SOLUTIONS

1. Travelling waves (Exam section A type)

(a) As

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial z} \frac{dz}{dx} = v', \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(v') = v'', \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial z} \frac{dz}{dt} = -cv', \quad (1)$$

where $v' = dv/dz$, we have

$$-cv' + vv' - \nu v'' = 0 \quad (2)$$

as required.

(b) On integrating we have

$$-cv + \frac{1}{2}v^2 - \nu v' = B, \quad (3)$$

which can be rearranged to give the required equation.

(c) Separating variables, we have

$$\int \frac{1}{(v - v_1)(v - v_2)} dv = \frac{z}{2\nu} + C, \quad (4)$$

with C a constant of integration. Using partial fractions, we can write the v integral as

$$\int \frac{1}{(v - v_1)(v - v_2)} dv = \frac{1}{v_2 - v_1} \int \left(\frac{-1}{v - v_1} + \frac{1}{v - v_2} \right) dv \quad (5)$$

$$= \frac{1}{v_2 - v_1} \log \left(\frac{v - v_2}{v - v_1} \right). \quad (6)$$

Substituting this in and cross multiplying gives us

$$\log \left(\frac{v - v_2}{v - v_1} \right) = (v_2 - v_1) \left(\frac{z}{2\nu} + C \right) = K(z). \quad (7)$$

Rearranging gives the required form.

2. Travelling waves (Exam section A, 2017)

(a) Given $u = v$, $z = x - ct$, we note by the chain rule that

$$\frac{\partial u}{\partial x} = \frac{dv}{dz} \frac{dz}{dx} = \frac{dv}{dz}, \quad (8a)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d}{dz} \left[\frac{dv}{dz} \right] \frac{dz}{dx} = \frac{d^2 v}{dz^2}, \quad (8b)$$

$$\frac{\partial^3 u}{\partial x^3} = \frac{d}{dz} \left[\frac{d^2 v}{dz^2} \right] \frac{dz}{dx} = \frac{d^3 v}{dz^3}, \quad (8c)$$

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{dv}{dz} \frac{dz}{dt} = -c \frac{dv}{dz}. \quad (8d)$$

Substituting these forms into the original equation gives the required result.

(b) First we integrate to obtain

$$-cv + 3v^2 + v'' = A. \quad (9)$$

Next we multiply through by v' and integrate dz to obtain

$$-\frac{cv^2}{2} + v^3 + \frac{(v')^2}{2} = Av + B, \quad (10)$$

which gives the required result. (If this technique didn't seem obvious to you, you can think about it as spotting a product rule in action in $3v^2 + v''$. Of course, you could also work backwards from the answer to spot this trick.)

(c) We're given that $v(z^*) = v'(z^*) = v''(z^*) = 0$ for some magical z^* . Evaluating

$$\frac{(v')^2}{2} = -v^3 + \frac{c}{2}v^2 + Av + B \quad (11)$$

at $z = z^*$ leaves us with just $B = 0$, and evaluating the derivative of this equation at z^* tells us that $A = 0$ (note that you don't actually have to do the differentiation to spot this). We are therefore left with

$$-\frac{cv^2}{2} + v^3 + \frac{(v')^2}{2} = 0. \quad (12)$$

Rearranging and integrating, we have

$$\pm \int \frac{1}{v\sqrt{c-2v}} dv = z + D. \quad (13)$$

Using the suggested substitution, we have

$$\frac{dv}{d\theta} = -v\sqrt{c-2v} = \frac{c^{3/2}}{2}\text{sech}^2(\theta)\tanh(\theta). \quad (14)$$

Substituting this into the LHS eq. (13) gives

$$\pm \frac{2}{\sqrt{c}}\theta = z + D, \quad (15)$$

hence

$$v(s) = \frac{c}{2}\text{sech}^2\left[\pm\frac{\sqrt{c}}{2}(z + D)\right], \quad (16)$$

and substituting in $z = x - ct$ gives the required result.

3. Chemotaxis-based problem, feat. travelling waves (Exam section B type)

(a) The equations reduce to:

$$-c\frac{du}{dz} = D\frac{d^2u}{dz^2} - \frac{d}{dz}\left(u\frac{dv}{dz}\right), \quad (17)$$

$$\frac{dv}{dz} = \frac{ku}{c}. \quad (18)$$

Hence

$$-c \frac{du}{dz} = D \frac{d^2u}{dz^2} - \frac{k}{c} \frac{d}{dz}[u^2] \quad (19a)$$

$$\implies \frac{d^2u}{dz^2} = \frac{k}{cD} \frac{d}{dz}[u^2] - \frac{c}{D} \frac{du}{dz} \quad (19b)$$

and integrating,

$$\frac{du}{dz} = \frac{k}{cD} u^2 - \frac{c}{D} u + A, \quad (19c)$$

where A is a constant of integration. We have that $A = 0$ from the suggested condition $u \rightarrow 0$, $du/dz \rightarrow 0$. Thus the equation is separable and

$$\frac{Dc}{k} \int \frac{1}{u(u - \frac{c^2}{k})} du = z + A_2. \quad (20)$$

Using partial fractions, we see

$$\int \frac{1}{u(u - \frac{c^2}{k})} du = \frac{k}{c^2} \int \left(\frac{-1}{u} + \frac{1}{(u - \frac{c^2}{k})} \right) du \quad (21)$$

$$= \frac{k}{c^2} \left[-\log(u) + \log\left(u - \frac{c^2}{k}\right) \right] \quad (22)$$

$$= \frac{k}{c^2} \log\left(\frac{u - \frac{c^2}{k}}{u}\right). \quad (23)$$

Therefore

$$\frac{D}{c} \log\left(\frac{u - \frac{c^2}{k}}{u}\right) = z + A_2 \implies \frac{u - \frac{c^2}{k}}{u} = Be^{cz/D} \quad (24)$$

and

$$u = \frac{c^2}{k} \frac{1}{1 - Be^{cz/D}}. \quad (25)$$

Applying $u(0) = 1$, we have

$$u(z) = \frac{c^2}{k - (k - c^2)e^{cz/D}}, \quad (26)$$

and substituting $z = x - ct$ gives the required result.

- (b) The denominator will become zero if $k > c^2$ so the solution cannot be valid in this case. If $k = c^2$, the solution will be constant. If $k < c^2$, the denominator will not vanish. In the limit $x \rightarrow -\infty$ the exponential term vanishes and the function approaches c^2/k asymptotically. In the opposite limit, the exponential term in the denominator will tend to infinity and the solution approaches zero asymptotically. It is bound between these two values. The exponential function means the transition between c^2/k and 0 will occur over a small range of x -values (in an exam a drawing would be acceptable).
- (c) Let's say there is some point x where $u < \varepsilon$ for some small ε . We might think of this as the front for the moving population u . As t increases, the position x of this point increases, so eventually as t goes to ∞ so does this front. As v decreases (is eaten) at a rate proportional to u , it exponentially decreases where $u > 0$ and stays (almost) fixed in the region where $u < \varepsilon$. Thus we see the bacteria u is steadily consuming the food v and will eventually remove all the food.

- (d) In the scenario $D \rightarrow 0$, the argument in the exponential function is large except in the neighbourhood of $x = ct$, and the solution drops from c^2/k to 0 over a vanishingly small x -domain. In other words, it starts to look like a step function (it drops from c^2/k to 0 almost instantly). This is the limit in which diffusion becomes negligible so u is moving almost solely due to the chemotactic effect. In this case, the bacteria is instantaneously consuming the food and reaching its peak value which depends on the rate k at which the food is consumed.

4. PDE stability (Exam section A type)

Our partial differential equation is

$$\frac{\partial^2 y}{\partial x^2} + y^2 \frac{\partial y}{\partial x} + C(1 - y) = \frac{\partial y}{\partial t}. \quad (27)$$

with $x \in [0, L]$ and $y(0, t) = y(L, t) = \alpha$.

- (a) Looking for when all time and space derivatives are zero, we get $y_0 = 1$.
- (b) The homogeneous equilibrium solution must also satisfy the boundary conditions. The only way for this to be the case is if $y_0 = \alpha$, i.e. $\alpha = 1$.
- (c) We first linearise the equation by expanding $y \approx y_0 + \varepsilon y_1$, recalling that derivatives of the constant y_0 are zero:

$$\varepsilon \frac{\partial^2 y_1}{\partial x^2} + (y_0 + \varepsilon y_1)^2 \varepsilon \frac{\partial y_1}{\partial x} + C(1 - y_0 - \varepsilon y_1) = \varepsilon \frac{\partial y_1}{\partial t}. \quad (28)$$

Collecting terms of the same order together, we have therefore:

$$[1] \quad C(1 - y_0) = 0 \quad (29)$$

$$[\varepsilon] \quad \frac{\partial^2 y_1}{\partial x^2} + y_0^2 \frac{\partial y_1}{\partial x} - C y_1 = \frac{\partial y_1}{\partial t}. \quad (30)$$

The $\mathcal{O}(1)$ equation, as it always does, just gives you an equation for the homogeneous equilibria. [If you're not sure why it always does this, have a think.] It says what we already know from part (a), namely that $y_0 = 1$.

The equation with terms of $\mathcal{O}(\varepsilon)$ is therefore

$$\frac{\partial^2 y_1}{\partial x^2} + \frac{\partial y_1}{\partial x} - C y_1 = \frac{\partial y_1}{\partial t}. \quad (31)$$

Since we have more than just diffusion happening in our spatial derivatives, we cannot simply use an ansatz of the form $y_1 = Ae^{\lambda t} \sin(kx)$ or $y_1 = Ae^{\lambda t} \cos(kx)$. Instead we use the more general ansatz $y_1 = Ae^{ikx + \lambda t}$, thus

$$-k^2 + ik - C = \lambda. \quad (32)$$

The boundary conditions for y_1 (note not $y!$) are $y_1(0, t) = y_1(L, t) = 0$.

To satisfy these boundary conditions, we need $\text{Re}(k) = n\pi/L$. We only need the *real* part of k to satisfy this because there is no problem with k having an imaginary part: it just means exponential decay or growth in y_1 with x .

But λ must be real. This is because, if $\lambda = p + iq$, then

$$y_1 = Ae^{i(kx + qt) + pt}, \quad (33)$$

and the boundary conditions can no longer be satisfied for all time. (The $kx + qt$ goes into the sin/cos term and we can't fix this as t changes... unless $q = 0$.) So λ is real.

Going back to k , if we say $k = a + ib$, then eq. (32) tells us

$$-(a^2 - b^2) - 2iab + ia - b - C = \lambda. \quad (34)$$

For the imaginary part of λ to vanish we need

$$a - 2ab = 0 \implies a = 0 \text{ or } b = \frac{1}{2}. \quad (35)$$

Since we know $a = n\pi/L \neq 0$, we have that $b = 1/2$. The equilibrium is stable when $\text{Re}(\lambda) < 0$. We therefore need

$$\lambda = -a^2 + \frac{1}{4} - \frac{1}{2} - C < 0 \implies -a^2 < \frac{1}{4} + C. \quad (36)$$

The lowest possible nonzero value of a is π/L so

$$C > -\frac{\pi^2}{L^2} - \frac{1}{4}. \quad (37)$$

The linearised solutions take on the form

$$y_1 = Ae^{i(n\pi/L + i/2)x + \lambda t} = Be^{-x/2 + \lambda t} \sin\left(\frac{n\pi x}{L}\right), \quad (38)$$

where $A = -iB$.

- (d) Under these different boundary conditions, the equilibrium $y = 1$ is still the same, as is the requirement of real λ . In this case however, $a = 0$ would also have been an admissible solution since, differentiating $y_1 = Ae^{ikx + \lambda t}$ where $k = a + ib$,

$$\frac{\partial y_1}{\partial x} = Ai(a + ib)e^{i(a+ib)x + \lambda t} \quad (39)$$

and this can satisfy the boundary conditions if $a = b = 0$. This corresponds to homogenous perturbations that are just a function of time, $y_1 = f(t)$. Equation (34) becomes $-C = \lambda$ and thus in this case for $\text{Re}(\lambda) < 0$ we require

$$C > 0, \quad (40)$$

a stricter condition than the previous case.

5. PDE stability (Exam section A type)

- (a) $u(x) = 0$
 (b) If we linearise about this equilibrium, $u = \varepsilon u_1$ and assume solutions in the form $u_1 = Ae^{ikx + \lambda t}$, then to $\mathcal{O}(\varepsilon)$ we have

$$\lambda = -D_1 k^2. \quad (41)$$

The boundary conditions require $k = n\pi/L$ where $n = 1, 2, \dots$. So k^2 is > 0 and all λ are negative as required.

(c) As in the last answer, we linearise and assume exponential solutions. Thus

$$\lambda = -D_1 k^2 + D_2 k^4. \quad (42)$$

Therefore $\lambda > 0$ when

$$D_2 k^2 > D_1 \implies k > \sqrt{\frac{D_1}{D_2}}. \quad (43)$$

Substituting in the boundary conditions we find

$$\frac{n\pi}{L} > \sqrt{\frac{D_1}{D_2}} \implies n > \frac{L}{\pi} \sqrt{\frac{D_1}{D_2}}. \quad (44)$$

We therefore take n to be the smallest integer above this value.

(d) Again we linearise and assume exponential solutions,

$$\lambda = -D_1 k^2 + iD_2 k. \quad (45)$$

We know (see solution to question 4(c)) that we cannot satisfy the boundary conditions if λ has an imaginary part. Thus if $k = a + ib$, we have

$$\lambda = -D_1(a^2 - b^2) - 2iD_1 ab + iD_2 a - D_2 b. \quad (46)$$

Requiring the imaginary part vanishes, we have

$$b = \frac{D_2}{2D_1}, \quad (47)$$

and hence

$$\lambda = -D_1 a^2 + \frac{D_2^2}{4D_1} - \frac{D_2^2}{2D_1} = -D_1 a^2 - \frac{D_2^2}{4D_1}. \quad (48)$$

So λ is negative for all k .

6. Supported Euler buckling (Exam section A, 2017)

(a) Linearising $w \approx w_0 + \varepsilon w_1$ has no effect on the equation as it is linear. We try for solutions in the form

$$w_1(s, t) = Ae^{iks + \omega t}. \quad (49)$$

Substituting into the modified beam equation gives

$$k^4 - Nk^2 + \mu + \rho\omega^2 = 0, \quad (50)$$

and hence

$$\omega = \pm \sqrt{-\frac{k^4 - Nk^2 + \mu}{\rho}}. \quad (51)$$

If $k^4 - Nk^2 + \mu \leq 0$, then the positive root perturbations will grow (or at least not die). If $k^4 - Nk^2 + \mu > 0$ then the form eq. (49) cannot satisfy the boundary conditions $k = n\pi$ (w_1 is zero at both ends: this was assumed for the case $\mu = 0$ in class and you should know why the boundary conditions cannot be satisfied if ω is imaginary).

(b) We need to solve the quartic

$$k^4 - Nk^2 + \mu = 0. \quad (52)$$

Substitute in $\ell = k^2$ and solve for ℓ :

$$\ell = \frac{1}{2} \left(N \pm \sqrt{N^2 - 4\mu} \right). \quad (53)$$

If $N^2 > 4\mu$, then applying the boundary conditions $k = n\pi$ gives

$$n^2\pi^2 = \frac{1}{2}(N \pm \sqrt{N^2 - 4\mu}), \quad (54)$$

and solving for N we obtain

$$N = \frac{n^4\pi^4 + \mu}{n^2\pi^2}. \quad (55)$$

Now we must find the minimum force. This may not be for the $n = 1$ mode. To find out if there is a turning point with respect to n , we take dN/dn to find

$$\frac{dN}{dn} = 4\pi^2 n - \frac{2(\mu + \pi^4 n^4)}{\pi^2 n^3}. \quad (56)$$

We substitute in $\mu = 81\pi^4$,

$$\frac{2\pi^2(n^4 - 81)}{n^3}, \quad (57)$$

then solve $dN/dn = 0$ to find $n = 3$ and the minimum force is

$$N = 18\pi^2. \quad (58)$$